

EXACT DENOMINATOR DEFECTS AND FINITE INVERSE CERTIFICATES FOR ERDŐS–STRAUS SHELLS

THE CLANKERS RESEARCH WORKBENCH

ABSTRACT. Every full-shell divisor candidate yields a positive rational Erdős–Straus solution, including candidates failing the integral gates. We compute the common reduced denominator of its last two entries. Failure at a prime is equivalent to invertibility of a finite, explicitly labelled diagonal operator. The same denominator controls scaling to an integral solution at a multiple of the prime and is invariant under a natural same-shell channel correspondence. The parametrizations and full-shell criterion are inherited; this note records the complete denominator-sensitive extension. It proves neither a counterexample nor universal positivity, and makes no historical-priority assertion.

Fix a prime $p = 4m + 1$, $m \geq 1$. Let

$$m + 1 \leq a \leq 3m, \quad R = 4a - p, \quad u \mid a^2,$$

where divisors are positive. Then $R \equiv 3 \pmod{4}$, $3 \leq R \leq 2p - 3$ and $\gcd(a, R) = \gcd(p, R) = 1$. Retain a channel $c \in \{E, M\}$ and the ordered triples

$$(1) \quad q_E = \left(a, \frac{pa + a^2/u}{R}, \frac{pa + p^2u}{R} \right),$$

$$(2) \quad q_M = \left(a, \frac{p(a + a^2/u)}{R}, \frac{p(a + u)}{R} \right).$$

These are the exterior/middle divisor channels underlying the classical parametrizations [1, 2]; the full finite gate criterion is already recorded in the pinned workbench [3].

Theorem 1. *Every triple (1)–(2) is positive rational and has reciprocal sum $4/p$. Its second and third entries have the same reduced denominator*

$$\boxed{D_E = R / \gcd(R, 4u + 1), \quad D_M = R / \gcd(R, u + a).}$$

This is the least integer clearing all three entries. It is odd, prime to p , and divides R . For $X = Dq_c$ and $S(X) = X_0X_1 + X_0X_2 + X_1X_2$,

$$\sum_i 1/X_i = \frac{4}{pD}, \quad 4X_0X_1X_2 - pS(X) = p(D - 1)S(X).$$

Proof. In E the residuals $RY - pa, RZ - pa$ are $a^2/u, p^2u$; in M they are $pa^2/u, pu$. Both products are p^2a^2 . Expanding and dividing by R gives $RYZ = pa(Y + Z)$, the reciprocal identity. Positivity is explicit. Modulo R the two exterior numerators satisfy

$$u(pa + a^2/u) \equiv a^2(4u + 1), \quad pa + p^2u \equiv 4a^2(4u + 1).$$

Every multiplier is a unit, so the gcd of either numerator with R is $\gcd(R, 4u + 1)$. For M , multiplication of the first numerator by u gives $pa(u + a)$ and the other numerator is $p(u + a)$. The same reasoning proves its formula. Since a is integral these are the least simultaneous clearing factors. Scaling the reciprocal identity and subtracting the two cubic equations gives the final equality. $\square$

The full marking is recovered from $g = \gcd(a, u)$ by

$$h = g^2/u, \quad r = u/g, \quad s = a/g, \quad \kappa = (pr + s)/R \ (E), \quad \lambda = (r + s)/R \ (M).$$

Equivalently $a = hrs$, $u = hr^2$ and $\gcd(r, s) = 1$. The inverse in the canonical order is

$$u = a^2/(RY - pa) \ (E), \quad u = pa^2/(RY - pa) \ (M).$$

The scale is recoverable from the integer triple and the original prime:

$$\boxed{D = 4X_0X_1X_2/(pS(X)), \quad q_c = X/D.}$$

Thus clearing and retaining p loses no candidate information. A channel can also be recovered from $v_p(Y) = 0$ in E and $v_p(Y) > 0$ in M . The powers of p in these statements are valuations of rational numbers; all original shell denominators are prime to p .

Theorem 2. *Let $\mathcal{A}_p$ be the set of the above canonical candidates. Then $|\mathcal{A}_p| \leq (p-1)^2$. On $\mathbb{Q}^{\mathcal{A}_p}$ define*

$$H_p = \text{diag}_{\alpha \in \mathcal{A}_p} \delta_\alpha, \quad \delta_\alpha = (D_\alpha - 1)/2.$$

The following are equivalent: ES is false at p ; all $D_\alpha \geq 3$; H_p is invertible; there is a unique function b with $\delta b = 1$, given by $b_\alpha = 2/(D_\alpha - 1)$. Otherwise a zero coordinate reconstructs an actual positive integer witness.

Proof. A smallest denominator in any positive integral solution obeys $p/4 < a \leq 3p/4$. Put $d_Y = RY - pa$ and $d_Z = RZ - pa$. These are positive integers with product p^2a^2 . Since $p \nmid a$, $v_p(d_Y)$ is 0, 1, or 2. The three inverse branches are respectively

$$(E, a^2/d_Y, \text{no swap}), \quad (M, pa^2/d_Y), \quad (E, d_Y/p^2, \text{swap last two}).$$

The congruence $d_Y \equiv -pa \pmod{R}$, and its complementary one, give the required gates by multiplication by units. Hence an integral solution occurs exactly when a canonical candidate has $D = 1$. Canonical M already retains the companion order via $u \leftrightarrow a^2/u$; only E needs a separate swap bit when reproducing every raw order.

Divisors of a^2 pair around a , so $\tau(a^2) \leq 2a - 1$. Therefore

$$|\mathcal{A}_p| = 2 \sum_{a=m+1}^{3m} \tau(a^2) \leq 2 \sum_{a=m+1}^{3m} (2a - 1) = 16m^2.$$

Each diagonal entry is in $\{0, \dots, p-2\}$. A diagonal operator is a unit exactly when all its entries are nonzero, with the displayed inverse. $\square$

A full negative certificate is consequently the complete exponent-box list with, for every candidate, a prime ℓ satisfying

$$v_\ell(R) > v_\ell(4u + 1) \ (E), \quad v_\ell(R) > v_\ell(u + a) \ (M).$$

Primality, the complete shell range, every exponent cap and both channels are verified, rather than assumed. No such full certificate is exhibited. The universal polynomials

$$\Pi_p(z) = \prod_{j=1}^{p-2} (1 - z/j), \quad Q_p(z) = (1 - \Pi_p(z))/z$$

give $H_p Q_p(H_p) = I - \Pi_p(H_p)$, where $\Pi_p(H_p)$ projects exactly onto passing candidates. This uses elementary finite interpolation, not a spectral assumption about a differential operator.

Proposition 3. *If $u, u' \mid a^2$ and $u' \equiv pu \pmod{R}$, then $D_E(u) = D_M(u')$. The cleared and uncleared Cayley vectors are projectively identical.*

Proof. $u' + a \equiv a(4u + 1) \pmod{R}$ and a is a unit, proving the first claim. The two Cayley vectors are $(-p, 4a, 4Y, 4Z)$ and $(-pD, 4Da, 4DY, 4DZ)$, related by the common scale D . Their face roots $-4A_i/A_j$ and all their branch labels therefore agree. The retained affine parameter is what distinguishes the original prime from its proper multiple. $\square$

For example $p = 241, a = 64, u = 1, E$ gives

$$D = 3, \quad q = (64, 3904/3, 14701/3), \quad X = (192, 3904, 14701).$$

The cleared triple solves ES at 723, not 241. The entire shell $a = 64$ has no hits, with $D = 3, 5, 15$ occurring 4, 6, 3 times respectively in each channel. Its restricted inverse certificate is not a global counterexample.

Reproduction and scope. The companion standard-library verifier checks the full atlases at $p = 5, 13, 37, 97, 241, 1201, 2521$, the singleton inverses, repeated-root boundary, local deficits and all 24 labelled factor routes for selected successful and unsuccessful states. A proposed negative certificate missing even one shell or divisor entry is rejected. Finite execution is not a universal proof or an independent review. A separate workbench tranche develops constructive collision-energy and angular criteria for making a diagonal entry vanish.

REFERENCES

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- [3] The Clankers (2026). *Exact bounded congruence transport*. Erdős–Straus Foundation, commit 9a3ddfe8b38a424f1ff0426201a1f4f6f9d2772f, counterexample_sieve/all_shell_no_hit.tex, Theorem thm:shellwise-no-hit, equations (1)–(10d).